Blueline tilefish (*Caulolatilus microps*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	2.84	1.56	
Complexity in Reproductive Strategy	2.08	1.36	
Dispersal of Early Life Stages	1.84	2	
Egg and Larval Survival and Settlement Requirements	2.28	1.06	
Historical Variability in Temperature	2.52	1.54	
Juvenile Habitat Requirements	1.16	2.4	
Other Stressors	1.68	1.7	
Population Growth Rate	2.76	2.24	
Prey Specificity	1.76	2.06	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	2.08	2.2	
Spawning Cycle	2	2.5	
Stock Size/Status	1.96	1.9	
Total Sensitivity | Certainty	**Moderate (2) | 0.9**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	1.05	0.25	
Bottom Oxygen	1.47	0.70	
Bottom Salinity	3.58	0.69	
Bottom Aragonite Solubility	3.31	1.01	
Net primary production	1.1	0.18	
Downward Flux of Particulate Organic Carbon	1.07	0.10	
Total Exposure	**Moderate (2.35)**	**0.85**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (4.7)**	**1.7**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Blueline tilefish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High (Burton et al. 2023)	Moderate	Decrease
Exposure	Very High (Burton et al. 2023)	Moderate	Decrease
Vulnerability	Very High (Burton et al. 2023)	Moderate	Decrease
Directional Effect	Neutral (Burton et al. 2023)	Neutral	None

Stocks

Blueline tilefish is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

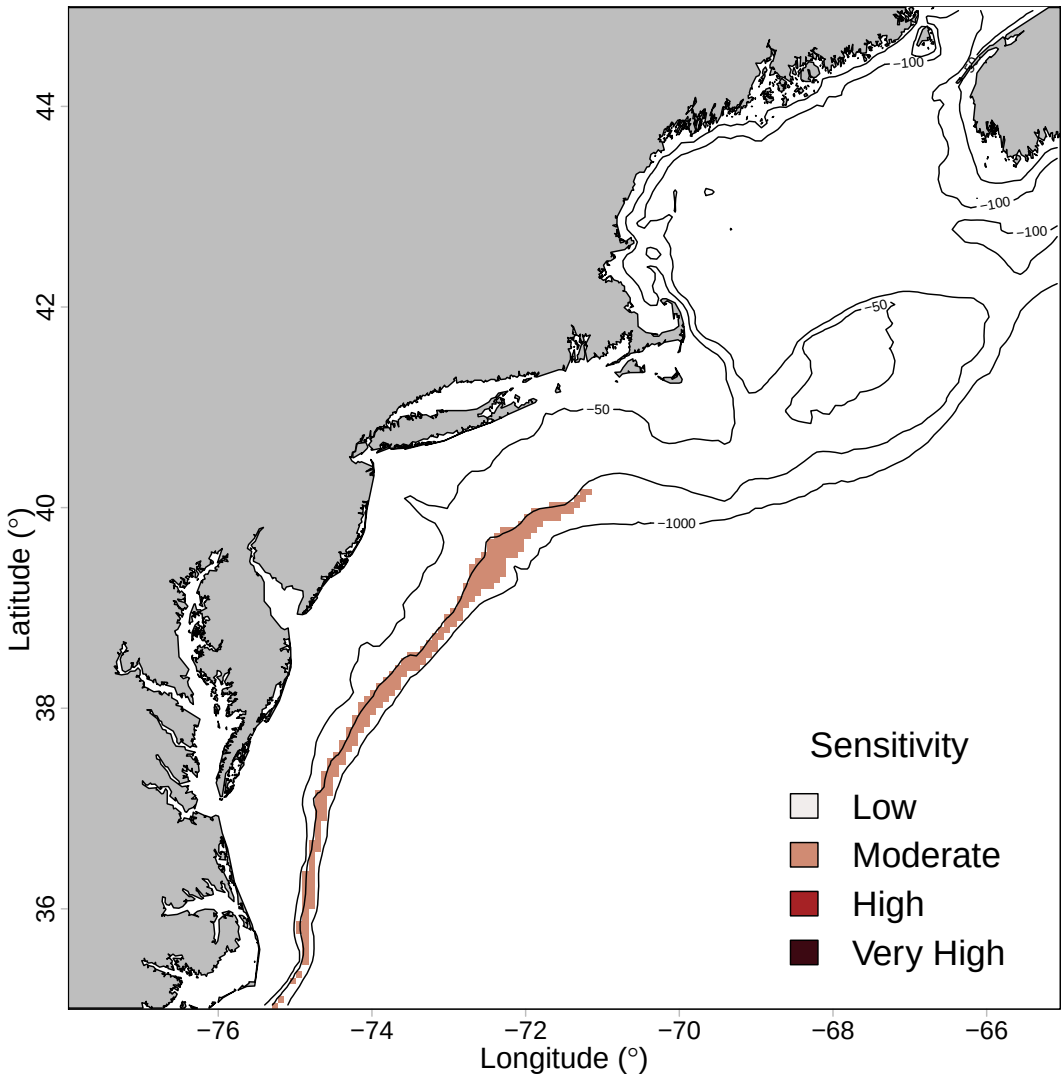


Figure 2: Blueline tilefish Sensitivity on the Northeast Shelf

Blueline tilefish received a **Moderate** sensitivity score. Certainty was 90% (Figure 1). Three attributes had scores greater than 2.5: Adult Mobility (2.8), Population Growth Rate (2.8), and Historical Variability in Temperature (2.5) (Figure 1).

Adult blueline tilefish are primarily demersal. They are thought to be sedentary and remain close to their dens, similar to goldenline tilefish (Schmidtke 2017). Maximum ages upwards of 50 years have been recorded (Schmidtke 2017) (Spanik and Ballenger 2023). They reach maturity anywhere from a few months to 3.7 years off the coast of Virginia to upwards of 5 years off the southeast US (Ross 1982) (Harris et al. 2004) (Low 2023).

Blueline tilefish were not included in the 2016 Climate Vulnerability Assessment for the Northeast US, but were included in the recent Climate Vulnerability Assessment for the Southeast US (Burton et al. 2023). Blueline tilefish received a High sensitivity score in that assessment. This score was driven by Sensitivity to Temperature and Population Growth Rate attributes. The experts in that assessment suggested that

long-lived, slow-growing nature of blueline tilefish are believed to make them susceptible to overfishing (Burton et al. 2023).

Exposure:

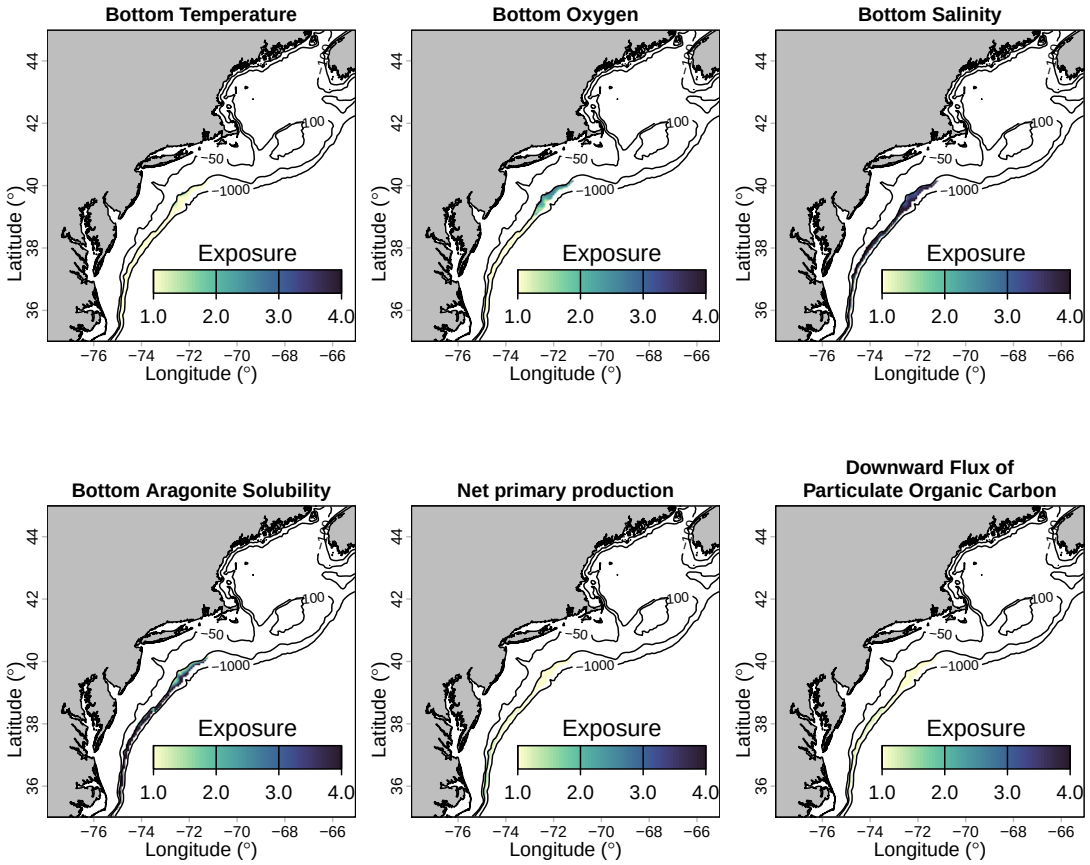


Figure 3: Maps of each Exposure Factor for Blueline tilefish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Blueline tilefish along the Northeast Shelf was **Moderate** (2.35 +/- 0.85) (Figure 1). Exposure was relatively consistent across the species distribution in this region and was driven by high bottom salinity and aragonite solubility exposure (Figure 4; Figure 3).

Blueline tilefish were not included in the previous assessment on the Northeast Shelf but were recently assessed in the Southeast US (Burton et al. 2023). In that assessment they received a Very High exposure score, which was driven by high sea surface temperature, salinity, ocean acidification, and current changes (Burton et al. 2023).

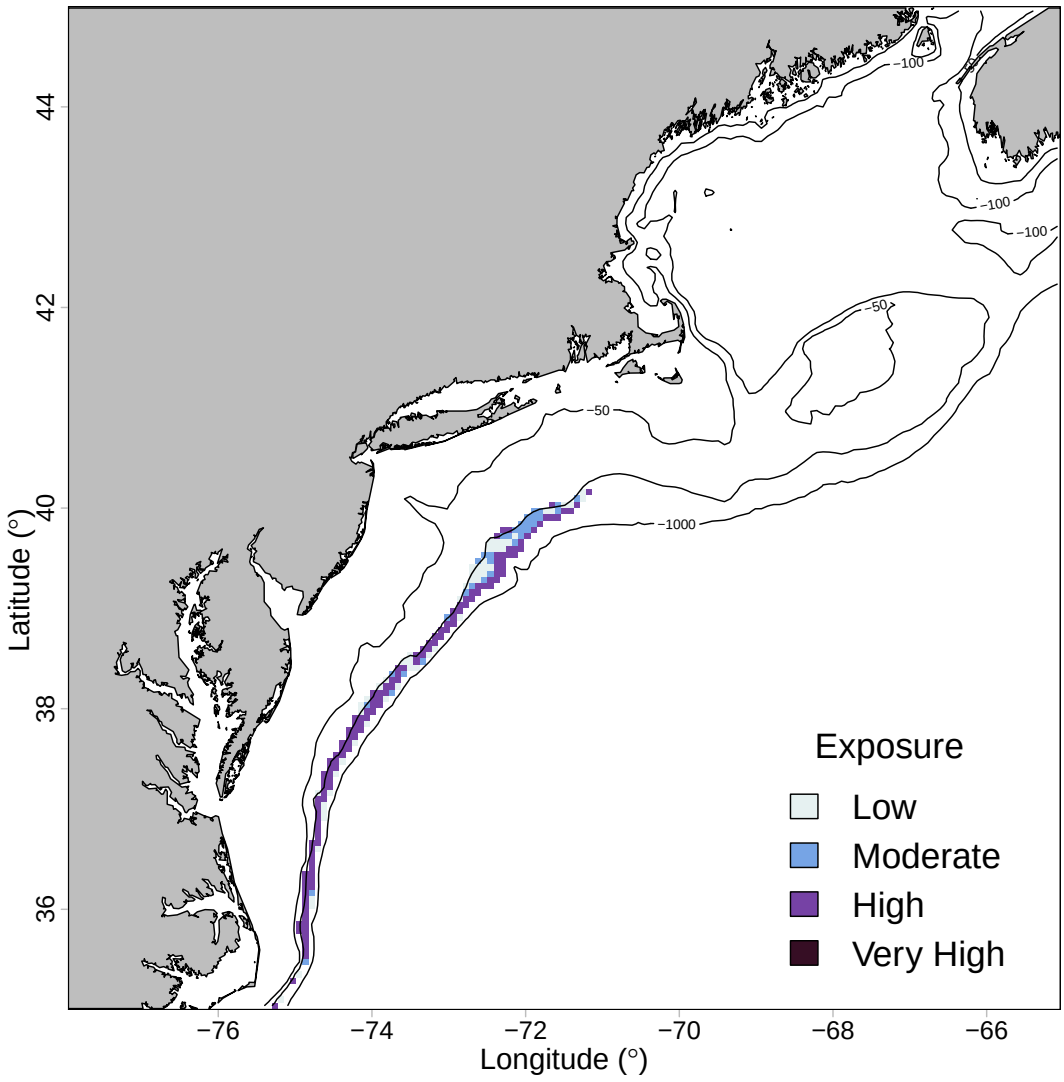


Figure 4: Total Exposure on the Northeast Shelf for Blueline tilefish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

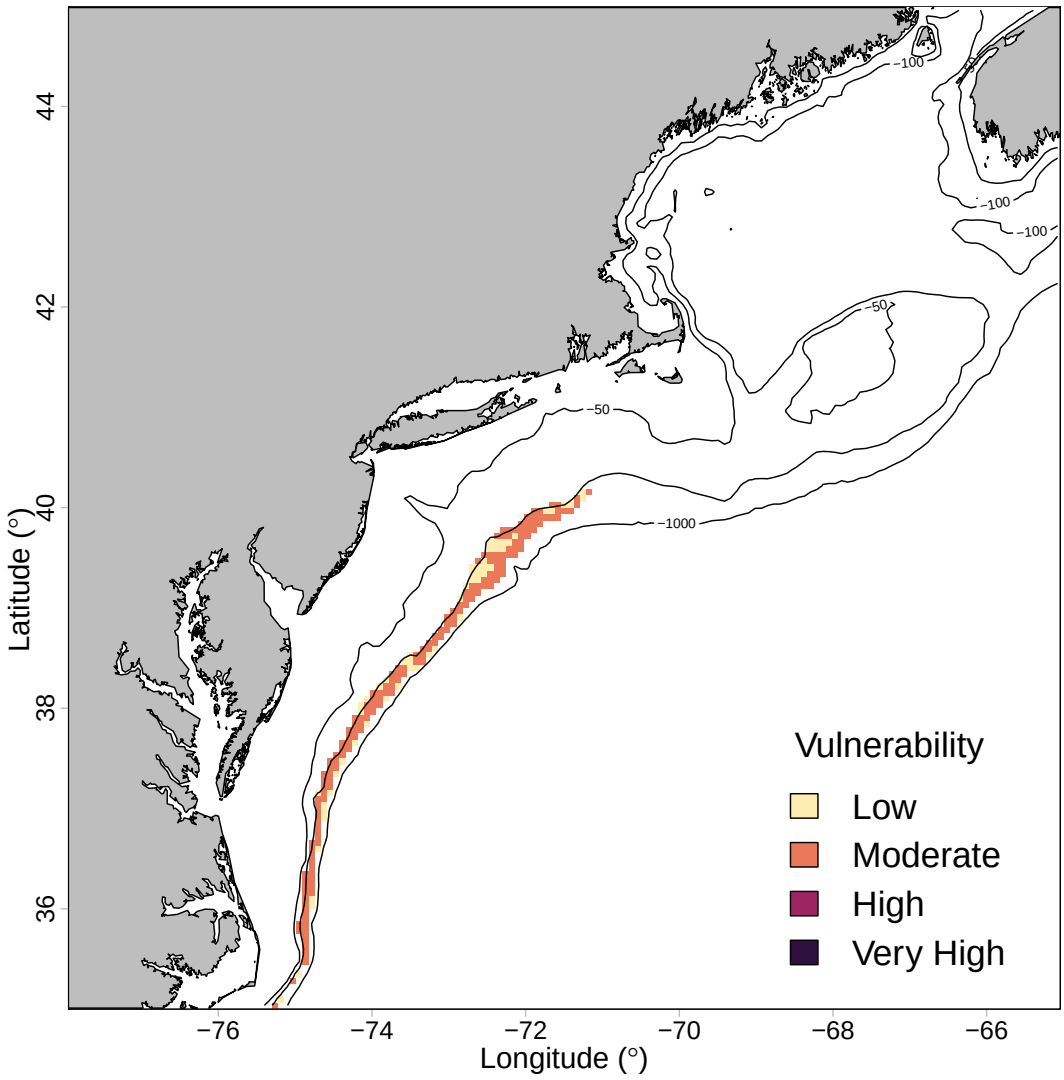


Figure 5: Vulnerability on the Northeast Shelf for Blueline tilefish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Blueline Tilefish throughout its range on the Northeast Shelf was **Moderate** (4.7 +/- 1.70) (Figure 1). Vulnerability was consistent throughout its range in the region (Figure 5).

In the previous climate vulnerability assessment for the Southeast US, blueline tilefish received a Very High Vulnerability score, driven by Very High exposure and High sensitivity scores (Burton et al. 2023). Blueline tilefish have not been assessed in the Northeast previously (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

Only 63% of data quality scores were greater than or equal to 2, indicating poor data quality. Most attributes had data gaps (mean attribute data quality score < 2), including Adult Mobility, Complexity in Reproductive Strategy, Egg and Larval Survival and Settlement Requirements, Historical Variability in Temperature, Other Stressors, and Stock Size/Status (Figure 1).

Model Confidence Score:

The mean model confidence score for blueline tilefish was 1.6 (+/- 1.02), indicating that experts had low to moderate confidence in the model. Scores ranged from 0 to 2.5. Experts said that the model captured the distribution of blueline tilefish well, but low sample size and poor sampling of rocky burrow habitats decreased the confidence. The model AUC score was 0.97, indicating excellent predictive performance.

Additional Metrics

Distribution Shifts:

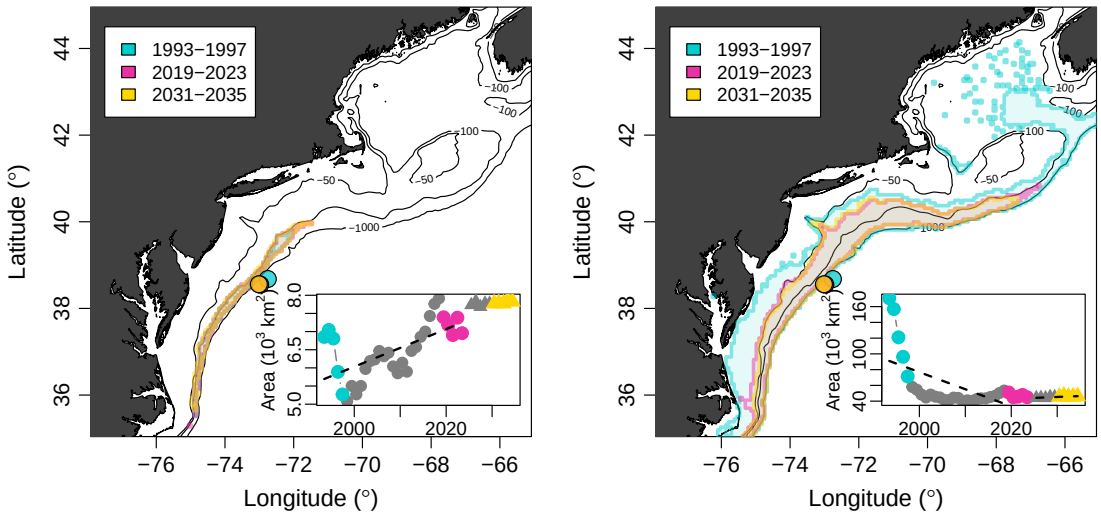


Figure 6: Distribution shifts on the Northeast Shelf for Blueline tilefish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Blueline tilefish experienced a significant loss in total habitat area between 1993 and 2023 due to loss of habitat in the Gulf of Maine and on the continental shelf off the coast of Maryland, Virginia, and North

Carolina. However, core habitat area on the continental shelf break grew significantly over the same time period (Figure 6). The distribution centroid shifted 24 km southwest.

Under near future climate conditions, blueline tilefish habitat is expected to remain similar (Figure 6). The distribution centroid is predicted to shift a further 4 km westward.

In the recent Climate Vulnerability Assessment for the Southeast Shelf, blueline tilefish received a moderate Distributional Vulnerability Rank due to limited adult mobility, high habitat specialization, and low larval dispersal (Burton et al. 2023).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	6,088	114,118	5.36	6.98
2019-2023	23.9 km SW	7,192 (+1,104)	47,239 (-66,879)	4.36 (-1.0)	5.29 (-1.68)
2031-2035	3.6 km W	7,823 (+630)	45,917 (-1,322)	4.42 (+0.06)	4.97 (-0.32)

Directional Effect:

The impact of climate change on blueline tilefish is expected to be **neutral**, with 92% certainty in expert scores. It is possible that blueline tilefish biomass may increase with increasing temperatures, as landings in the Mid-Atlantic have been correlated to increasing temperatures (Pinsky et al. 2018) (Rogers et al. 2019). However, little research has been conducted on how blueline tilefish will respond to climate change. The Climate Vulnerability Assessment for the Southeast US also predicted that the effect of climate change on blueline tilefish would be neutral (Burton et al. 2023).

Climate Effects on Abundance and Distribution

Little research has been conducted on the response of blueline tilefish to climate change. Rising water temperatures likely drove a rapid increase in landings in the Mid-Atlantic throughout the early 21st century as blueline tilefish distributions expanded north beyond their traditional South Atlantic boundary (Pinsky et al. 2018) (Rogers et al. 2019). Blueline tilefish may exhibit a moderately-high risk to achieving optimal biomass and yield due to climate change on productivity (Mid-Atlantic Fishery Management Council (MAFMC) 2025).

Life History Synopsis

Blueline tilefish have only recently been found on the Northeast U.S. Continental Shelf. They live in crevices or on ledges on the shelf, shelf break, and shelf slope from Georges Bank to Mexico and the Gulf of America (Schmidtke 2017). Blueline tilefish are multiple batch spawners and presumably spawn on/near the bottom (Low 2023). Spawning occurs from March through October in the southern portion range, and May - November in the Mid-Atlantic Bight (Harris et al. 2004) (Schmidtke 2017). Eggs and larvae are estimated to remain pelagic for 4-6 weeks and depend on currents for transport. Eddies and gyres may provide local retention while the Gulf Stream transports eggs and larvae up the East Coast. It seems

likely that substantial numbers of blueline larvae are transported out of the eastern Gulf of Mexico, to the South Atlantic, and from the South Atlantic into the Mid-Atlantic over the course of a few weeks (SouthEast Data, Assessment, and Review (SEDAR) 2025). Juveniles settle to the bottom and occupy benthic habitats around offshore submarine canyons on the shelf edge, constructing burrows in seafloor sediments or within rocks and boulders (Schmidtke 2017). Juveniles and adults consume a diverse array of benthic invertebrates, including polychaetes, crustaceans, and decapods. Gastropods and copepods have also been observed and are likely consumed when diel vertical migrations bring them close to the seafloor (Bielsa and Labisky 1987) (Ross 1982). Age at maturity is up to 3.7 years in the Mid-Atlantic and 5 years in the South Atlantic (Harris et al. 2004) (Low 2023). Maximum age is 37 years, although it is theorized that individuals could possibly reach upwards of 50 years in age (Schmidtke 2017) (Spanik and Ballenger 2023). In the Northwest Atlantic, blueline tilefish are managed by the Mid-Atlantic Fishery Management Council under the Tilefish Fishery Management Plan. Council management of blueline tilefish in the Mid-Atlantic began as a response to increased commercial and recreational landings in the region. The Mid-Atlantic stock has not been assessed. Population levels and fishing status are currently unknown but management measures have been in place since 2017 (Mid-Atlantic Fishery Management Council (MAFMC) 2017).

Supplemental Scores

Exposure Timeseries

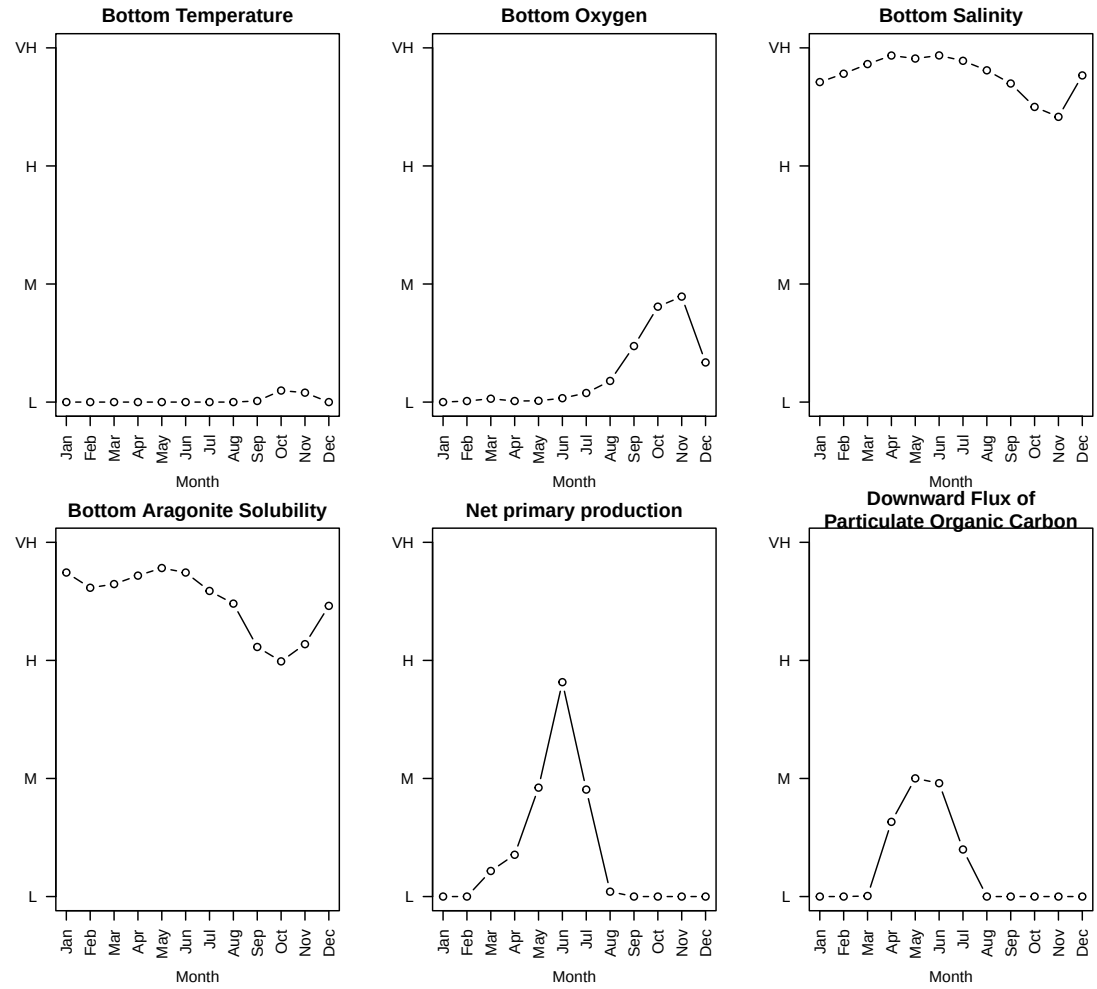


Figure 7: Monthly Timeseries of each Exposure Factor for Blueline tilefish. Exposure was calculated between 2014-2023 and 2025-2035

There was little month-to-month variation in exposure scores for blueline tilefish. Scores were consistently High, except in October where they decreased to Moderate (Figure 8). This was due to declines in bottom aragonite solubility (Figure 7).

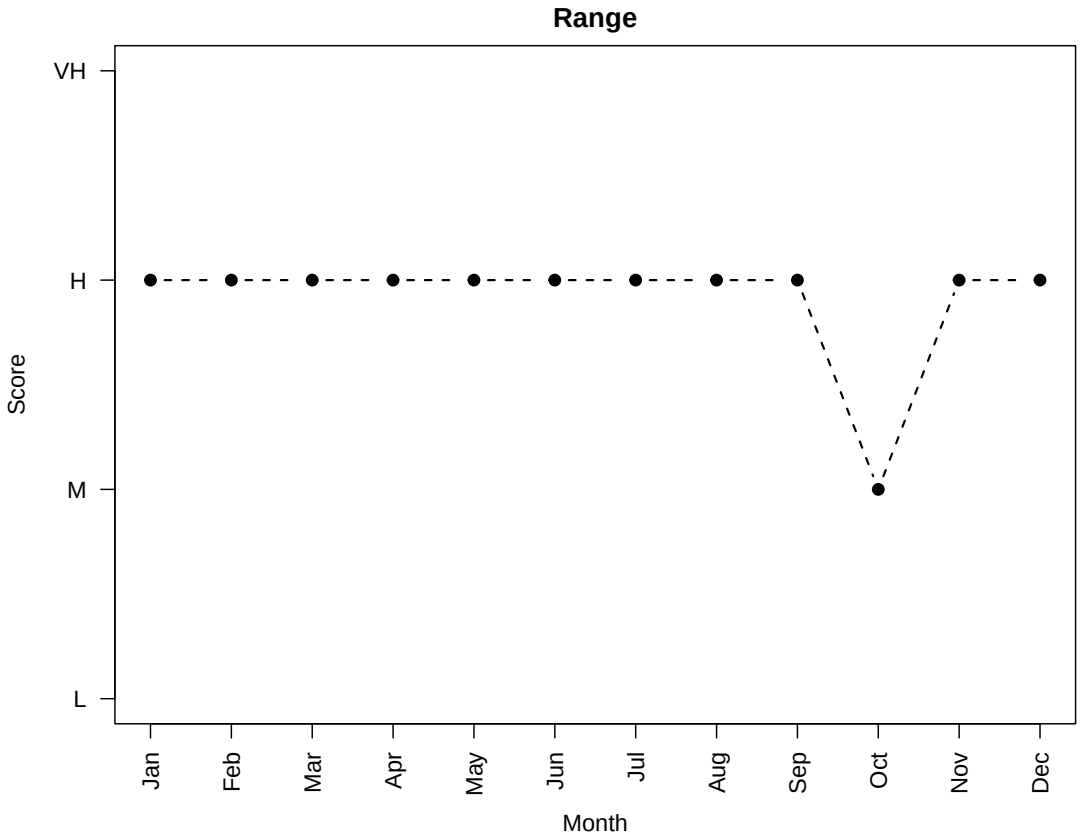


Figure 8: Timeseries of Total Exposure for Blueline tilefish

Exposure due to Significant Factors

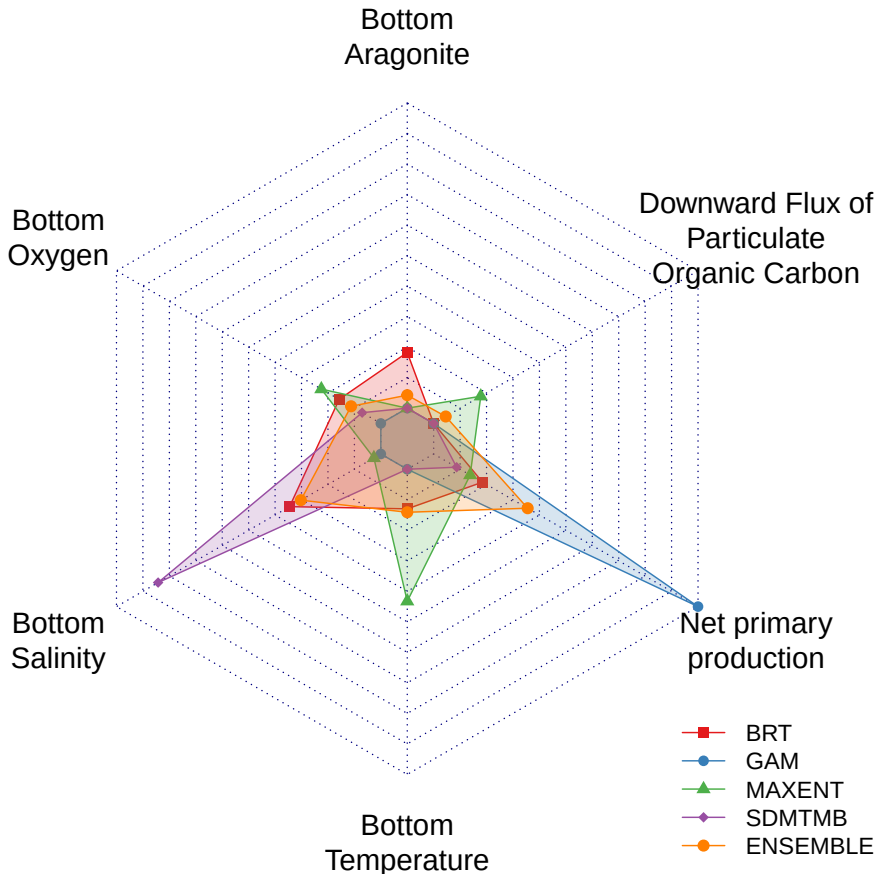


Figure 9: Dynamic Factors driving Blueline tilefish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, salinity, and net primary production were the major dynamic drivers of species distributions along the Northeast Shelf (Figure 9). Blueline tilefish are demersal species that occupy the shelfbreak and submarine canyons, so they prefer consistent conditions (Schmidtke 2017). Blueline tilefish are generalist predators, consuming polychaetes, crustaceans, decapods, and zooplankton (Bielsa and Labisky 1987) (Ross 1982). When exposure was calculated using only these factors, exposure decreased to Low due to low temperature, oxygen, and net primary productivity exposures (Figure 10; Figure 11).

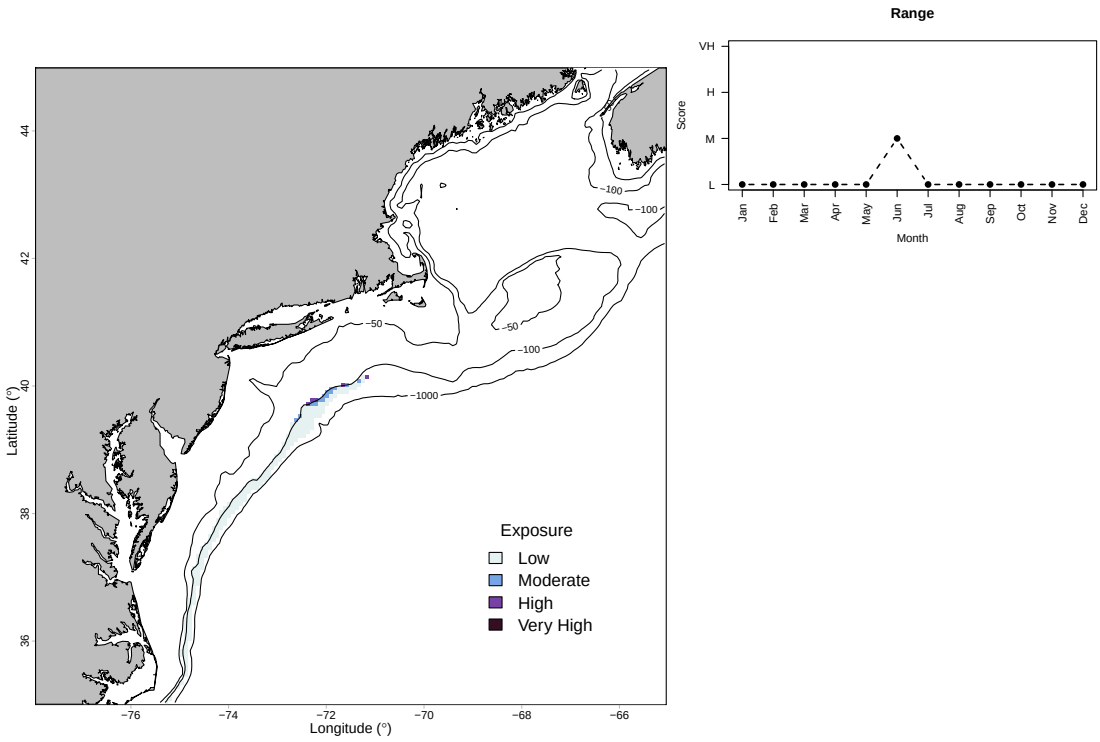


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Blueline tilefish distributions

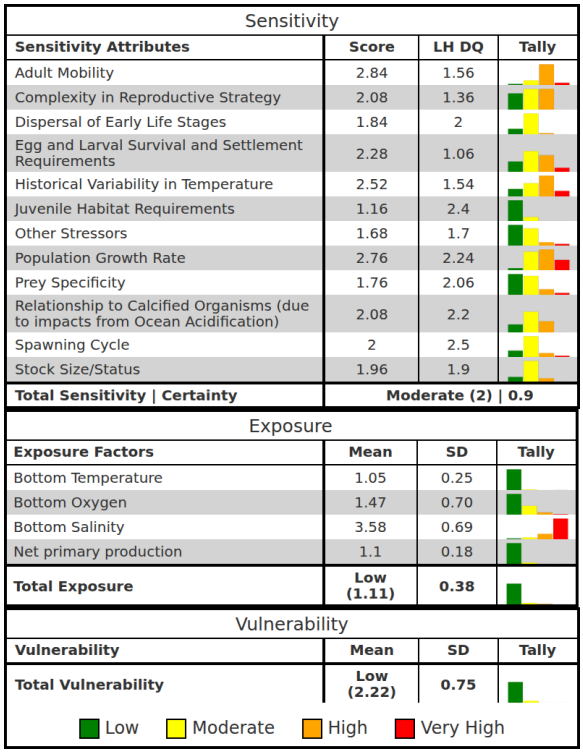


Figure 11: As in Figure 1 for Blueline tilefish, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

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